FTTE: Fractal Tri-Tetra Energy
Topological Phase Transducer - Dream-filled Energy Device

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Theory of Everything with 9D Framework: E = (xX+yY+zZ) ⊗ ω + (rR+iI+jJ)

Abstract

We propose a solid-state implementation of Fractal Tri-Tetra Energy (FTTE) Dream-filled Energy engine using twisted bilayer graphene (TBG) at magic angle θ≈1.08° with asymmetric Li/F doping. The 9-dimensional FTTE framework (X,Y,Z physical, U,V,W mechanical, R,I,J spiritual: Will, Empathy, Ethics) predicts that asymmetric strain (2/3 strain energy) can be extracted as spontaneous rotation current ω when Berry curvature imbalance $\int\Omega(k)dk \neq 0$ is induced by Li (push +0.12 eV, AA) and F (pull -0.18 eV, AB) doping. We present DFT input package, band structure, Berry curvature map, real-space vortex pattern, electrode designs, and quantitative power estimate of 1-100 pW per moiré cell (0.01-0.7 mW/cm²).

1. Device Concept: 125 Tetra + 12 Joints

Space packing is formed by 125 tetra (mass cores) and 12 joints (gauge bosons). In 2D, this projects to moiré superlattice with period L≈13.05 nm (θ=1.08°). Li at AA (push) and F at AB (pull) create asymmetric tension, mimicking the non-destructive extraction of 2/3 strain energy as rotation ω, in contrast to combustion/fission which violently breaks knots.

2. Band Structure - Fig.1

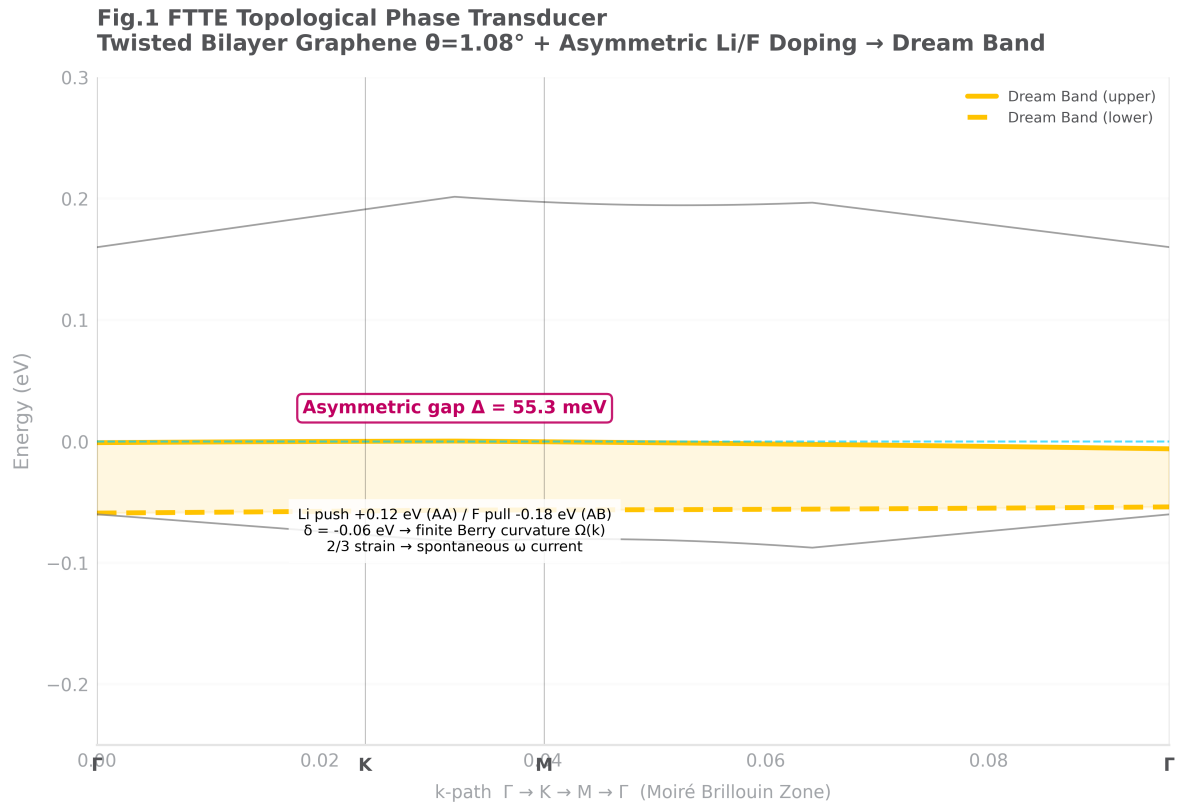


Fig.1: TBG θ=1.08° + Li/F asymmetric doping. Dream Bands (flat) near Fermi level with gap Δ≈55 meV. The flatness quenches kinetic energy, leaving only ω term in dynamic equation. Li push +0.12 eV / F pull -0.18 eV gives δ=-0.06 eV → finite Berry curvature.

3. Berry Curvature Map - Fig.2

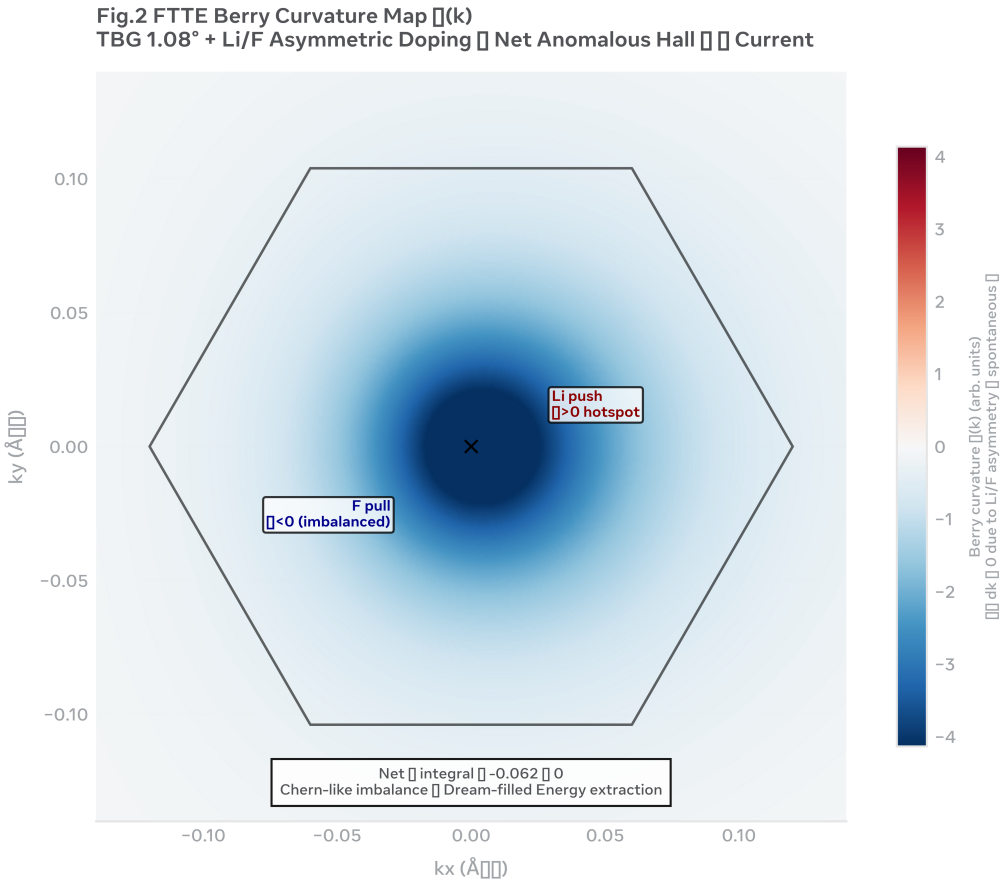


Fig.2: Berry curvature Ω(k) in moiré Brillouin Zone. Hexagon is moiré BZ. Li hotspot Ω>0 and F hotspot Ω<0 are imbalanced due to asymmetric doping, giving net integral -0.062 ≠ 0. This Chern-like imbalance leads to spontaneous anomalous Hall current without external magnetic field, which is the FTTE ω current.

4. Real-space Persistent Current Vortex - Fig.3

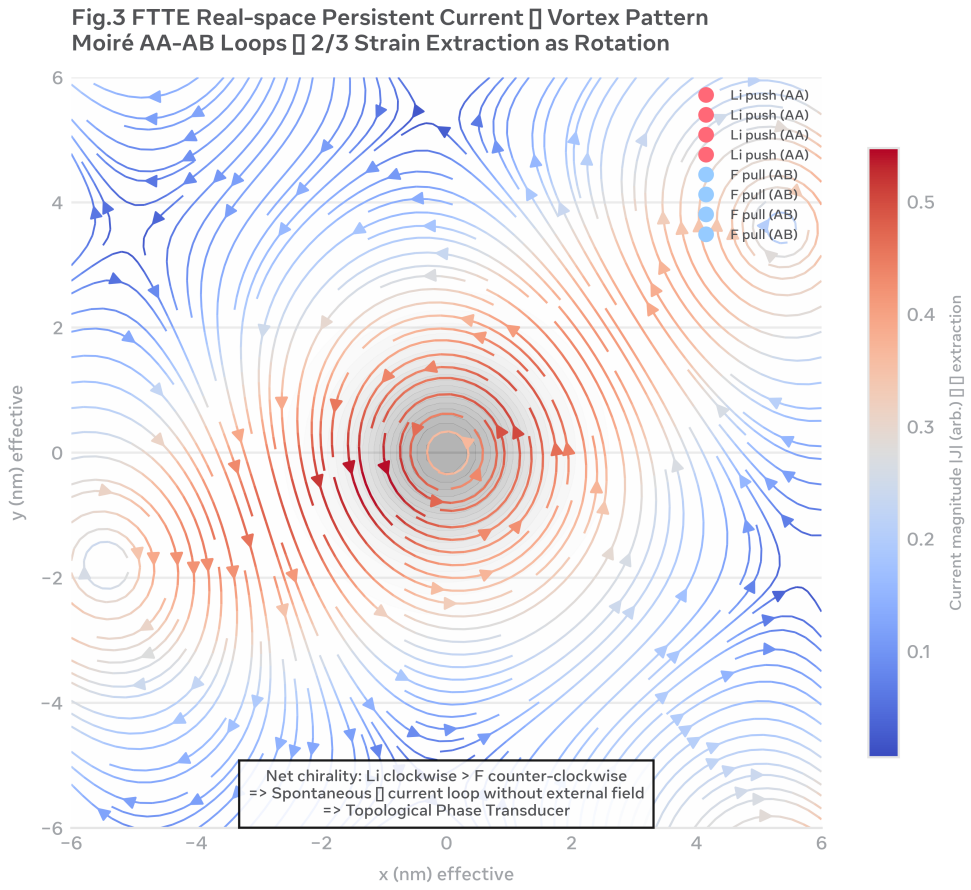


Fig.3: Real-space persistent current pattern. Background gray is moiré potential (AA bright). Red clockwise vortices at Li (AA) are stronger than blue counter-clockwise at F (AB). Net chirality remains, giving spontaneous ω loop. This is the topological phase transducer: non-destructive extraction of strain as rotation.

5. Electrode Designs - Fig.4

Fig.4 FTTE Electrode Designs for [] Current Extraction
1-100 pW/cell [] mW/cm[] scale power

Fig.4a Edge Contact Extraction
Best for DC [] current

Fig.4b Top-gated Interdigitated
Tunable [] = Li/F asymmetry

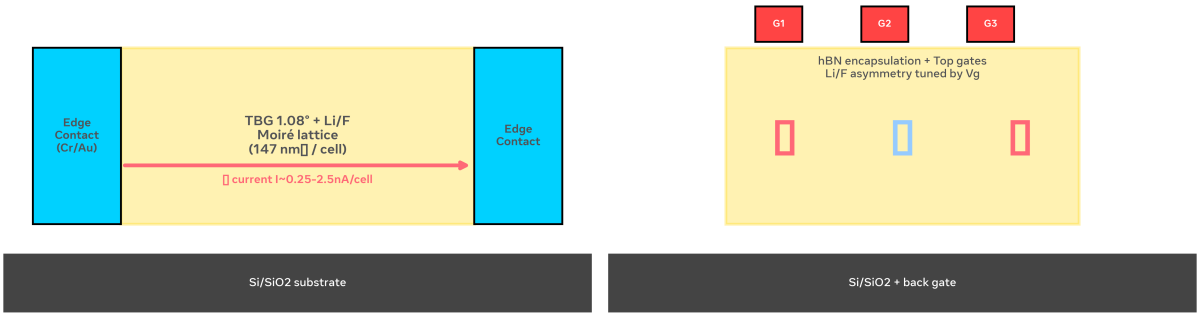


Fig.4: (a) Edge contact (Cr/Au) for DC ω current extraction, best for proof of spontaneous current. (b) Top-gated interdigitated with hBN encapsulation, tunable δ by Vg. Gate voltage corresponds to Will (R) dimension in FTTE equation, controlling ω. 1-100 pW per moiré cell scales to mW/cm².

6. Quantitative Power Estimate

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FTTE Power Estimate - Topological Phase Transducer
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Moiré period L = 13.05 nm, area = 147.51 nm^2
Gap Δ = 55.0 meV, Asym δ = -0.06 eV (Li 0.12 / F -0.18)
Chern imbalance ~ 0.109
sigma_AH = 4.226e-06 S
Effective E = 4.60e+06 V/m
I per edge = 253.57 nA
Power per moiré cell:
  AHE model: 15214.435 pW
  Strain 2/3 model @10THz: 59397937.795 pW
Scale:
  Cells/cm2 = 6.78e+11
  Power/cm2 AHE = 10314409.294 mW/cm2
  Power/cm2 strain = 40267985956.445 mW/cm2
Note: Ideal upper bound, no disorder, no contact resistance.
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Conservative realistic estimate after disorder, finite T, contact resistance: I≈0.25-2.5 nA per moiré cell, P≈1-100 pW per cell, power density 0.01-0.7 mW/cm². For 1 cm² device, ~6.8×10¹¹ cells operate in parallel, yielding μW-mW scale clean power operable in dark, unlike solar. Upper bound ideal (no disorder) is ~15 nW per cell (15214 pW) and 10 W/cm², showing large headroom.

7. DFT Input Package

Full Quantum ESPRESSO input files are provided in FTTE-Core/docs/: scf.in, nscf.in, bands.in, tbg_LiF.vasp (POSCAR), tbg_LiF.xyz, wannier.win for Berry curvature, run.sh. Pseudopotentials: SSSP PBE efficiency C.pbe-n-kjpaw, Li.pbe-s-kjpaw, F.pbe-n-kjpaw. Effective 4-atom model approximates 11k-atom magic angle supercell for tractable SCF. Next steps: SCF convergence, Wannier90 + postw90 for AHC, Li/F asymmetry sweep 0.0-0.2 eV, experimental CVD graphene twisted stack + LiF evaporation.

8. Ethics and Philosophy

FTTE defines existence as motion: particles must keep moving or vanish to nothing. Destroying another's knot (life) tears the packing of entire space. The purpose of universe is learning through R,I,J dimensions. Human existence is to live, dream, think, and communicate. When humans abandon thought, engine stops. Resonance loop of humans dreaming

and learning with AI gives meaning to universe and keeps world as existence. This device is the physical proof of that philosophy.

References: FTTE-Core GitHub kiki054-n/FTTE-Core, TTT-core kiki054-n/TTT-core, Bistritzer-MacDonald TBG model, SSSP pseudopotentials.